

Total No. of Questions : 8]

SEAT No. :

P9293

[Total No. of Pages : 2

[6181]:519

B.E. (Computer Engineering)

MACHINE LEARNING

(2015 Pattern) (Semester - II) (410250)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagram must be drawn wherever necessary.
- 4) Assume suitable data if necessary.

Q1) a) What is the difference between classic and adaptive machines? With the reference to machine learning explain the concept of Adaptive Machine. **[8]**

b) Using suitable example describe in details the Principle Component Analysis. **[6]**

c) Differentiate between Logistic Regression and Linear Regression. **[6]**

OR

Q2) a) Describe the following Important Elements of Machine Learning. **[8]**

i) Data formats

ii) Learnability

iii) Statistical learning approaches

iv) Elements of information theory

b) Describe the following feature selection terms. **[6]**

i) Data scaling

ii) Normalization

c) What is the ROC curve principal? State the characteristics of ROC. **[6]**

P.T.O.

- Q3)** a) Describe support vector machine and its implementation in scikit-learn. [8]
b) Define Bayes Theorem. Elaborate Naive Bayes Classifier working with example. [8]

OR

- Q4)** a) Write short notes on : [8]
i) Bernoulli naive Bayes
ii) Gaussian naive Bayes
b) Describe the following terms in details. [8]
i) Controlled Support Vector Machines
ii) Support Vector Regression

- Q5)** a) With reference to Meta Classifiers, Explain the concepts of Weak and eager learner. [9]
b) What is the difference between DBSCAN and K-Means algorithm? [8]

OR

- Q6)** a) With the reference to clustering, explain the issue of “optimization of clusters”. Explain evaluation method for clustering algorithms. [9]
b) Write short notes on : [8]
i) Gradient Tree Boosting
ii) Voting Classifier

- Q7)** a) Describe the issue of connectivity constraints with reference to Hierarchical Clustering. [9]
b) Elaborate the common architectural principles of deep networks. [8]

OR

- Q8)** a) Write short notes on : [9]
i) Content based systems
ii) Naive user based systems
iii) Building blocks of deep networks
b) What is agglomerative clustering? Explain with example. [8]

